Complex Analysis

Problem

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Complex Analysis

Chapter 2. Cauchy's Theorem and Its Applications

1. Here are some examples of analytic functions on the unit disc that cannot be extended analytically past the unit circle. The following definition is needed. Let

f be a function defined in the unit disc $\mathbb{D}$, with boundary circle C. A point w on C is said to be *regular* for f if there is an open neighborhood U of w and an analytic function g on U, so that $f = g$ on $\mathbb{D} \cap U$. A function f defined on $\mathbb{D}$ cannot be continued analytically past the unit circle if no point of C is regular for f.

(a) Let

$$f(z) = \sum_{n=0}^{\infty} z^{2^n} \qquad \text{for } |z| < 1.$$

Notice that the radius of convergence of the above series is 1. Show that f cannot be continued analytically past the unit disc. [Hint: Suppose $\theta = 2\pi p/2^k$, where p and k are positive integers. Let $z = re^{i\theta}$; then $|f(re^{i\theta})| \to \infty$ as $r \to 1$.]

(b) * Fix $0 < \alpha < \infty$. Show that the analytic function f defined by

$$f(z) = \sum_{n=0}^{\infty} 2^{-n\alpha} z^{2^n} \qquad \text{for } |z| < 1$$

extends continuously to the unit circle, but cannot be analytically continued past the unit circle. [Hint: There is a nowhere differentiable function lurking in the background. See Chapter 4 in Book I.]

2.* Let

$$F(z) = \sum_{n=1}^{\infty} d(n) z^n \qquad \text{for } |z| < 1$$

where $d(n)$ denotes the number of divisors of n. Observe that the radius of convergence of this series is 1. Verify the identity

$$\sum_{n=1}^{\infty} d(n) z^n = \sum_{n=1}^{\infty} \frac{z^n}{1 - z^n}.$$

Using this identity, show that if $z = r$ with $0 < r < 1$, then

$$|F(r)| \geq \ c \frac{1}{1-r} \log(1/(1-r))$$

as $r \to 1$. Similarly, if $\theta = 2\pi p/q$ where p and q are positive integers and $z = re^{i\theta}$, then

$$|F(re^{i\theta})| \geq \ c_{p/q} \frac{1}{1-r} \log(1/(1-r))$$

as $r \to 1$. Conclude that F cannot be continued analytically past the unit disc.

3. Morera's theorem states that if f is continuous in $\mathbb{C}$, and $\int_T f(z)\,dz = 0$ for all triangles T, then f is holomorphic in $\mathbb{C}$. Naturally, we may ask if the conclusion still holds if we replace triangles by other sets.

(a) Suppose that f is continuous on $\mathbb{C}$, and

$$\int_C f(z)\,dz = 0 \tag{16}$$

for every circle C. Prove that f is holomorphic.

(b) More generally, let Γ be any toy contour, and $\mathcal{F}$ the collection of all translates and dilates of Γ. Show that if f is continuous on $\mathbb{C}$, and

$$\int_\gamma f(z)\,dz = 0 \qquad \text{for all } \gamma \in \mathcal{F}$$

then f is holomorphic. In particular, Morera's theorem holds under the weaker assumption that $\int_T f(z)\,dz = 0$ for all equilateral triangles.

[Hint: As a first step, assume that f is twice real differentiable, and write $f(z) = f(z_0) + a(z - z_0) + b(\overline{z - z_0}) + O(|z - z_0|^2)$ for z near z_0. Integrating this expansion over small circles around z_0 yields $\partial f/\partial \overline{z} = b = 0$ at z_0. Alternatively, suppose only that f is differentiable and apply Green's theorem to conclude that the real and imaginary parts of f satisfy the Cauchy-Riemann equations.

In general, let $\varphi(w) = \varphi(x, y)$ (when $w = x + iy$) denote a smooth function with $0 \le \varphi(w) \le 1$, and $\int_{\mathbb{R}^2} \varphi(w)\,dV(w) = 1$, where $dV(w) = dxdy$, and $\int$ denotes the usual integral of a function of two variables in $\mathbb{R}^2$. For each $\epsilon > 0$, let $\varphi_\epsilon(z) = \epsilon^{-2}\varphi(\epsilon^{-1}z)$, as well as

$$f_\epsilon(z) = \int_{\mathbb{R}^2} f(z - w)\varphi_\epsilon(w)\,dV(w),$$

where the integral denotes the usual integral of functions of two variables, with $dV(w)$ the area element of $\mathbb{R}^2$. Then f_ϵ is smooth, satisfies condition (16), and $f_\epsilon \to f$ uniformly on any compact subset of $\mathbb{C}$.]

4. Prove the converse to Runge's theorem: if K is a compact set whose complement if not connected, then there exists a function f holomorphic in a neighborhood of K which cannot be approximated uniformly by polynomial on K.

[Hint: Pick a point z_0 in a bounded component of K^c, and let $f(z) = 1/(z - z_0)$. If f can be approximated uniformly by polynomials on K, show that there exists a polynomial p such that $|(z - z_0)p(z) - 1| < 1$. Use the maximum modulus principle (Chapter 3) to show that this inequality continues to hold for all z in the component of K^c that contains z_0.]

5.* There exists an entire function F with the following "universal" property: given any entire function h, there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{n\to\infty} F(z+N_k) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$.

(a) Let $p_1, p_2, \ldots$ denote an enumeration of the collection of polynomials whose coefficients have rational real and imaginary parts. Show that it suffices to find an entire function F and an increasing sequence $\{M_n\}$ of positive integers, such that

$$|F(z) - p_n(z-M_n)| < \frac{1}{n} \qquad \text{whenever } z \in D_n, \tag{17}$$

where D_n denotes the disc centered at M_n and of radius n. [Hint: Given h entire, there exists a sequence $\{n_k\}$ such that $\lim_{k\to\infty} p_{n_k}(z) = h(z)$ uniformly on every compact subset of $\mathbb{C}$.]

(b) Construct F satisfying (17) as an infinite series

$$F(z) = \sum_{n=1}^{\infty} u_n(z)$$

where $u_n(z) = p_n(z-M_n)e^{-c_n(z-M_n)^2}$, and the quantities $c_n > 0$ and $M_n > 0$ are chosen appropriately with $c_n \to 0$ and $M_n \to \infty$. [Hint: The function e^{-z^2} vanishes rapidly as $|z|\to\infty$ in the sectors $\{|\arg z| < \pi/4 - \delta\}$ and $\{|\pi - \arg z| < \pi/4 - \delta\}$.]

In the same spirit, there exists an alternate "universal" entire function G with the following property: given any entire function h, there is an increasing sequence $\{N_k\}_{k=1}^{\infty}$ of positive integers, so that

$$\lim_{k\to\infty} D^{N_k} G(z) = h(z)$$

uniformly on every compact subset of $\mathbb{C}$. Here $D^j G$ denotes the jth (complex) derivative of G.

Chapter 3. Meromorphic Functions and the Logarithm

1.* Consider a holomorphic map on the unit disc $f : \mathbb{D} \to \mathbb{C}$ which satisfies $f(0) = 0$. By the open mapping theorem, the image $f(\mathbb{D})$ contains a small disc centered at the origin. We then ask: does there exist $r > 0$ such that for *all* $f : \mathbb{D} \to \mathbb{C}$ with $f(0) = 0$, we have $D_r(0) \subset f(\mathbb{D})$?

(a) Show that with no further restrictions on f, no such r exists. It suffices to find a sequence of functions $\{f_n\}$ holomorphic in $\mathbb{D}$ such that $1/n \notin f(\mathbb{D})$. Compute $f_n'(0)$, and discuss.

(b) Assume in addition that f also satisfies $f'(0) = 1$. Show that despite this new assumption, there exists no $r > 0$ satisfying the desired condition.

[Hint: Try $f_\epsilon(z) = \epsilon(e^{z/\epsilon} - 1)$.]

The Koebe-Bieberbach theorem states that if in addition to $f(0) = 0$ and $f'(0) = 1$ we also assume that f is injective, then such an r exists and the best possible value is $r = 1/4$.

(c) As a first step, show that if $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is analytic and injective for $0 < |z| < 1$, then $\sum_{n=1}^{\infty} n|c_n|^2 \leq 1$.

[Hint: Calculate the area of the complement of $h(D_\rho(0) - \{0\})$ where $0 < \rho < 1$, and let $\rho \to 1$.]

(d) If $f(z) = z + a_2 z^2 + \cdots$ satisfies the hypotheses of the theorem, show that there exists another function g satisfying the hypotheses of the theorem such that $g^2(z) = f(z^2)$.

[Hint: $f(z)/z$ is nowhere vanishing so there exists ψ such that $\psi^2(z) = f(z)/z$ and $\psi(0) = 1$. Check that $g(z) = z\psi(z^2)$ is injective.]

(e) With the notation of the previous part, show that $|a_2| \leq 2$, and that equality holds if and only if

$$f(z) = \frac{z}{(1 - e^{i\theta} z)^2} \qquad \text{for some } \theta \in \mathbb{R}.$$

[Hint: What is the power series expansion of $1/g(z)$? Use part (c).]

(f) If $h(z) = \frac{1}{z} + c_0 + c_1 z + c_2 z^2 + \cdots$ is injective on $\mathbb{D}$ and avoids the values z_1 and z_2, show that $|z_1 - z_2| \leq 4$.

[Hint: Look at the second coefficient in the power series expansion of $1/(h(z) - z_j)$.]

(g) Complete the proof of the theorem. [Hint: If f avoids w, then $1/f$ avoids 0 and $1/w$.]

2. Let u be a harmonic function in the unit disc that is continuous on its closure. Deduce Poisson's integral formula

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} \frac{1 - |\, z_0|^2}{|e^{i\theta} - z_0|^2} u(e^{i\theta})\, d\theta \qquad \text{for } |z_0| < 1$$

from the special case $z_0 = 0$ (the mean value theorem). Show that if $z_0 = re^{i\varphi}$, then

$$\frac{1 - |\, z_0|^2}{|e^{i\theta} - z_0|^2} = \frac{1 - r^2}{1 - 2r\cos(\theta - \varphi) + r^2} = P_r(\theta - \varphi),$$

and we recover the expression for the Poisson kernel derived in the exercises of the previous chapter.

[Hint: Set $u_0(z) = u(T(z))$ where

$$T(z) = \frac{z_0 - z}{1 - \overline{z_0} z}.$$

Prove that u_0 is harmonic. Then apply the mean value theorem to u_0, and make a change of variables in the integral.]

3. If $f(z)$ is holomorphic in the deleted neighborhood $\{0 < |z - z_0| < r\}$ and has a pole of order k at z_0, then we can write

$$f(z) = \frac{a_{-k}}{(z - z_0)^k} + \cdots + \frac{a_{-1}}{(z - z_0)} + g(z)$$

where g is holomorphic in the disc $\{|z - z_0| < r\}$. There is a generalization of this expansion that holds even if z_0 is an essential singularity. This is a special case of the **Laurent series expansion**, which is valid in an even more general setting.

Let f be holomorphic in a region containing the annulus $\{z : r_1 \leq |\ z - z_0| \leq\ r_2\}$ where $0 < r_1 < r_2$. Then,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where the series converges absolutely in the interior of the annulus. To prove this, it suffices to write

$$f(z) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(\zeta)}{\zeta - z}\, d\zeta - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(\zeta)}{\zeta - z}\, d\zeta$$

when $r_1 < |z - z_0| < r_2$, and argue as in the proof of Theorem 4.4, Chapter 2. Here C_{r_1} and C_{r_2} are the circles bounding the annulus.

4.* Suppose Ω is a bounded region. Let L be a (two-way infinite) line that intersects Ω. Assume that $\Omega \cap L$ is an interval I. Choosing an orientation for L, we can define Ω_l and Ω_r to be the subregions of Ω lying strictly to the left or right of L, with $\Omega = \Omega_l \cup I \cup \Omega_r$ a disjoint union. If Ω_l and Ω_r are simply connected, then Ω is simply connected.

5.* Let

$$g(z) = \frac{1}{2\pi i} \int_{-M}^{M} \frac{h(x)}{x - z}\, dx$$

where h is continuous and supported in $[-M, M]$.

(a) Prove that the function g is holomorphic in $\mathbb{C} - [-M, M]$, and vanishes at infinity, that is, $\lim_{|z|\to\infty} |g(z)| = 0$. Moreover, the "jump" of g across $[-M, M]$ is h, that is,

$$h(x) = \lim_{\epsilon\to 0, \epsilon>0} g(x + i\epsilon) - g(x - i\epsilon).$$

[Hint: Express the difference $g(x + i\epsilon) - g(x - i\epsilon)$ in terms of a convolution of h with the Poisson kernel.]

(b) If h satisfies a mild smoothness condition, for instance a Hölder condition with exponent α, that is, $|h(x) - h(y)| \leq C|x - y|^{\alpha}$ for some $C > 0$ and all $x, y \in [-M, M]$, then $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly to functions $g_+(x)$ and $g_-(x)$ as $\epsilon \to 0$. Then, g can be characterized as the unique holomorphic function that satisfies:

(i) g is holomorphic outside $[-M, M]$,

(ii) g vanishes at infinity,

(iii) $g(x + i\epsilon)$ and $g(x - i\epsilon)$ converge uniformly as $\epsilon \to 0$ to functions $g_+(x)$ and $g_-(x)$ with

$$g_+(x) - g_-(x) = h(x).$$

[Hint: If G is another function satisfying these conditions, $g - G$ is entire.]

Chapter 4. The Fourier Transform

1. Suppose $\hat{f}(\xi) = O(e^{-a|\xi|^p})$ as $|\xi| \to \infty$, for some $p > 1$. Then f is holomorphic for all z and satisfies the growth condition

$$|f(z)| \leq \ Ae^{a|z|^q}$$

where $1/p + 1/q = 1$.

Note that on the one hand, when $p \to \infty$ then $q \to 1$, and this limiting case can be interpreted as part of Theorem 3.3. On the other hand, when $p \to 1$ then $q \to \infty$, and this limiting case in a sense brings us back to Theorem 2.1.

[Hint: To prove the result, use the inequality $-\xi^p + \xi u \leq u^q$, which is valid when ξ and u are non-negative. To establish this inequality, examine separately the cases $\xi^p \geq \xi u$ and $\xi^p < \xi u$; note also that the functions $\xi = u^{q-1}$ and $u = \xi^{p-1}$ are inverses of each other because $(p-1)(q-1) = 1$.]

2. The problem is to solve the differential equation

$$a_n \frac{d^n}{dt^n} u(t) + a_{n-1} \frac{d^{n-1}}{dt^{n-1}} u(t) + \cdots + a_0 u(t) = f(t)\,,$$

where $a_0, a_1, \ldots, a_n$ are complex constants, and f is a given function. Here we suppose that f has bounded support and is smooth (say of class C^2).

(a) Let

$$\hat{f}(z) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i z t}\, dt.$$

Observe that $\hat{f}$ is an entire function, and using integration by parts show that

$$|\hat{f}(x+iy)| \leq \frac{A}{1+x^2}$$

if $|y| \leq a$ for any fixed $a \geq 0$.

(b) Write

$$P(z) = a_n (2\pi i z)^n + a_{n-1} (2\pi i z)^{n-1} + \cdots + a_0.$$

Find a real number c so that $P(z)$ does not vanish on the line

$$L = \{z : z = x + ic, \quad x \in \mathbb{R}\}.$$

(c) Set

$$u(t) = \int_L \frac{e^{2\pi i z t}}{P(z)} \hat{f}(z)\, dz.$$

Check that

$$\sum_{j=0}^{n} a_j \left(\frac{d}{dt}\right)^j u(t) = \int_L e^{2\pi i z t} \hat{f}(z)\, dz$$

and

$$\int_L e^{2\pi i z t} \hat{f}(z)\, dz = \int_{-\infty}^{\infty} e^{2\pi i x t} \hat{f}(x)\, dx.$$

Conclude by the Fourier inversion theorem that

$$\sum_{j=0}^{n} a_j \left(\frac{d}{dt}\right)^j u(t) = f(t).$$

Note that the solution u depends on the choice c.

3.* In this problem, we investigate the behavior of certain bounded holomorphic functions in an infinite strip. The particular result described here is sometimes called the three-lines lemma.

(a) Suppose $F(z)$ is holomorphic and bounded in the strip $0 < \mathrm{Im}(z) < 1$ and continuous on its closure. If $|F(z)| \leq 1$ on the boundary lines, then $|F(z)| \leq 1$ throughout the strip.

(b) For the more general F, let $\sup_{x\in\mathbb{R}} |F(x)| = M_0$ and $\sup_{x\in\mathbb{R}} |F(x+i)| = M_1$. Then,

$$\sup_{x\in\mathbb{R}} |F(x+iy)| \leq M_0^{1-y} M_1^y, \qquad \text{if } 0 \leq y \leq 1.$$

(c) As a consequence, prove that $\log \sup_{x\in\mathbb{R}} |F(x+iy)|$ is a convex function of y when $0 \leq y \leq 1$.

[Hint: For part (a), apply the maximum modulus principle to $F_\epsilon(z) = F(z)e^{-\epsilon z^2}$. For part (b), consider $M_0^{z-1} M_1^{-z} F(z)$.]

4.* There is a relation between the Paley-Wiener theorem and an earlier representation due to E. Borel.

(a) A function $f(z)$, holomorphic for all z, satisfies $|f(z)| \leq A_\epsilon e^{2\pi(M+\epsilon)|z|}$ for all ϵ if and only if it is representable in the form

$$f(z) = \int_C e^{2\pi i z w} g(w)\, dw$$

where g is holomorphic outside the circle of radius M centered at the origin, and g vanishes at infinity. Here C is any circle centered at the origin of radius larger than M. In fact, if $f(z) = \sum a_n z^n$, then $g(w) = \sum_{n=0}^{\infty} A_n w^{-n-1}$ with $a_n = A_n (2\pi i)^{n+1}/n!$.

(b) The connection with Theorem 3.3 is as follows. For these functions f (for which in addition f and $\hat{f}$ are of moderate decrease on the real axis), one can assert that the g above is holomorphic in the larger region, which consists of the slit plane $\mathbb{C} - [-M, M]$. Moreover, the relation between g and the Fourier transform $\hat{f}$ is

$$g(z) = \frac{1}{2\pi i} \int_{-M}^{M} \frac{\hat{f}(\xi)}{\xi - z}\, d\xi$$

so that $\hat{f}$ represents the jump of g across the segment $[-M, M]$; that is,

$$\hat{f}(x) = \lim_{\epsilon \to 0, \epsilon > 0} g(x + i\epsilon) - g(x - i\epsilon).$$

See Problem 5 in Chapter 3.

Chapter 5. Entire Functions

1. Prove that if f is holomorphic in the unit disc, bounded and not identically zero, and $z_1, z_2, \ldots, z_n, \ldots$ are its zeros ($|z_k| < 1$), then

$$\sum_n (1 - |\, z_n|) < \infty.$$

[Hint: Use Jensen's formula.]

2.* In this problem, we discuss Blaschke products, which are bounded analogues in the disc of the Weierstrass products for entire functions.

(a) Show that for $0 < |\alpha| < 1$ and $|z| \leq r < 1$ the inequality

$$\left| \frac{\alpha + |\alpha| z}{(1 - \overline{\alpha} z)\alpha} \right| \leq \frac{1 + r}{1 - r}$$

holds.

(b) Let $\{\alpha_n\}$ be a sequence in the unit disc such that $\alpha_n \neq 0$ for all n and

$$\sum_{n=1}^{\infty} (1 - |\alpha_n|) < \infty.$$

Note that this will be the case if $\{\alpha_n\}$ are the zeros of a bounded holomorphic function on the unit disc (see Problem 1). Show that the product

$$f(z) = \prod_{n=1}^{\infty} \frac{\alpha_n - z}{1 - \overline{\alpha}_n z} \frac{|\alpha_n|}{\alpha_n}$$

converges uniformly for $|z| \leq r < 1$, and defines a holomorphic function on the unit disc having precisely the zeros α_n and no other zeros. Show that $|f(z)| \leq 1$.

3.* Show that $\sum \dfrac{z^n}{(n!)^\alpha}$ is an entire function of order $1/\alpha$.

4.* Let $F(z) = \sum_{n=0}^{\infty} a_n z^n$ be an entire function of finite order. Then the growth order of F is intimately linked with the growth of the coefficients a_n as $n \to \infty$. In fact:

(a) Suppose $|F(z)| \leq Ae^{a|z|^\rho}$. Then

$$\limsup_{n\to\infty} |a_n|^{1/n} n^{1/\rho} < \infty. \tag{8}$$

(b) Conversely, if (8) holds, then $|F(z)| \leq A_\epsilon e^{a_\epsilon |z|^{\rho+\epsilon}}$, for every $\epsilon > 0$.

[Hint: To prove (a), use Cauchy's inequality

$$|a_n| \leq \frac{A}{r^n} e^{ar^\rho},$$

and the fact that the function $u^{-n}e^{u^\rho}$, $0 < u < \rho$, attains its minimum value $e^{n/\rho}(\rho/n)^{n/\rho}$ at $u = n^{1/\rho}/\rho^{1/\rho}$. Then, choose r in terms of n to achieve this minimum.

To establish (b), note that for $|z| = r$,

$$|F(z)| \leq \sum \frac{c^n r^n}{n^{n/\rho}} \leq \sum \frac{c^n r^n}{(n!)^{1/\rho}}$$

for some constant c, since $n^n \geq n!$. This yields a reduction to Problem 3.]

Chapter 6. The Gamma and Zeta Functions

1. This problem provides further estimates for ζ and ζ' near $\mathrm{Re}(s) = 1$.

(a) Use Proposition 2.5 and its corollary to prove

$$\zeta(s) = \sum_{1 \le n < N} n^{-s} - \frac{N^{s-1}}{s-1} + \sum_{n \ge N} \delta_n(s)$$

for every integer $N \ge 2$, whenever $\mathrm{Re}(s) > 0$.

(b) Show that $|\zeta(1+it)| = O(\log|t|)$, as $|t| \to \infty$ by using the previous result with $N =$greatest integer in $|t|$.

(c) The second conclusion of Proposition 2.7 can be similarly refined.

(d) Show that if $t \neq 0$ and t is fixed, then the partial sums of the series $\sum_{n=1}^{\infty} 1/n^{1+it}$ are bounded, but the series does not converge.

2.* Prove that for $\mathrm{Re}(s) > 0$

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\{x\}}{x^{s+1}}\,dx$$

where $\{x\}$ is the fractional part of x.

3.* If $Q(x) = \{x\} - 1/2$, then we can write the expression in the previous problem as

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s\int_1^\infty \frac{Q(x)}{x^{s+1}}\,dx.$$

Let us construct $Q_k(x)$ recursively so that

$$\int_0^1 Q_k(x)\,dx = 0, \qquad \frac{dQ_{k+1}}{dx} = Q_k(x), \qquad Q_0(x) = Q(x) \qquad \text{and} \qquad Q_k(x+1) = Q_k(x).$$

Then we can write

$$\zeta(s) = \frac{s}{s-1} - \frac{1}{2} - s\int_1^\infty \left(\frac{d^k}{dx^k}Q_k(x)\right) x^{-s-1}\,dx\,,$$

and a k-fold integration by parts gives the analytic continuation for $\zeta(s)$ when $\mathrm{Re}(s) > -k$.

4.* The functions Q_k in the previous problem are related to the Bernoulli polynomials $B_k(x)$ by the formula

$$Q_k(x) = \frac{B_{k+1}(x)}{(k+1)!} \qquad \text{for } 0 \le x \le 1.$$

Also, if k is a positive integer, then

$$2\zeta(2k) = (-1)^{k+1}\frac{(2\pi)^{2k}}{(2k)!}B_{2k},$$

where $B_k = B_k(0)$ are the Bernoulli numbers. For the definition of $B_k(x)$ and B_k see Chapter 3 in Book I.

Chapter 7. The Zeta Function and Prime Number Theorem

1. Let $F(s) = \sum_{n=1}^{\infty} a_n/n^s$, where $|a_n| \leq M$ for all n.

(a) Then

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |F(\sigma + it)|^2 \, dt = \sum_{n=1}^{\infty} \frac{|a_n|^2}{n^{2\sigma}} \qquad \text{if } \sigma > 1.$$

How is this reminiscent of the Parseval-Plancherel theorem? See e.g. Chapter 3 in Book I.

(b) Show as a consequence the uniqueness of Dirichlet series: If $F(s) = \sum_{n=1}^{\infty} a_n n^{-s}$, where the coefficients are assumed to satisfy $|a_n| \leq cn^k$ for some k, and $F(s) \equiv 0$, then $a_n = 0$ for all n.

Hint: For part (a) use the fact that

$$\frac{1}{2T} \int_{-T}^{T} (nm)^{-\sigma} n^{-it} m^{it} \, dt \to \begin{cases} n^{-2\sigma} & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases}$$

2.* One of the "explicit formulas" in the theory of primes is as follows: if ψ_1 is the integrated Tchebychev function considered in Section 2, then

$$\psi_1(x) = \frac{x^2}{2} - \sum_\rho \frac{x^\rho}{\rho(\rho+1)} - E(x)$$

where the sum is taken over all zeros ρ of the zeta function in the critical strip. The error term is given by $E(x) = c_1 x + c_0 + \sum_{k=1}^\infty x^{1-2k}/(2k(2k-1))$, where $c_1 = \zeta'(0)/\zeta(0)$ and $c_0 = \zeta'(-1)/\zeta(-1)$. Note that $\sum_\rho 1/|\rho|^{1+\epsilon} < \infty$ for every $\epsilon > 0$, because $(1-s)\zeta(s)$ has order of growth 1. (See Exercise 8.) Also, obviously $E(x) = O(x)$ as $x \to \infty$.

3.* Using the previous problem one can show that

$$\pi(x) - \mathrm{Li}(x) = O(x^{\alpha+\epsilon}) \qquad \text{as } x \to \infty$$

for every $\epsilon > 0$, where α is fixed and $1/2 \leq \alpha < 1$ if and only if $\zeta(s)$ has no zeros in the strip $\alpha < \mathrm{Re}(s) < 1$. The case $\alpha = 1/2$ corresponds to the Riemann hypothesis.

4.* One can combine ideas from the prime number theorem with the proof of Dirichlet's theorem about primes in arithmetic progression (given in Book I) to prove the following. Let q and ℓ be relatively prime integers. We consider the primes belonging to the arithmetic progression $\{qk + \ell\}_{k=1}^{\infty}$, and let $\pi_{q,\ell}(x)$ denote the number of such primes $\leq x$. Then one has

$$\pi_{q,\ell}(x) \sim \frac{x}{\varphi(q) \log x} \qquad \text{as } x \to \infty\,,$$

where $\varphi(q)$ denotes the number of positive integers less than q and relatively prime to q.

Chapter 8. Conformal Mappings

1. Let f be a complex-valued C^1 function defined in the neighborhood of a point z_0. There are several notions closely related to conformality at z_0. We say that f is **isogonal** at z_0 if whenever $\gamma(t)$ and $\eta(t)$ are two smooth curves with $\gamma(0) = \eta(0) = z_0$, that make an angle θ there ($|\theta| < \pi$), then $f(\gamma(t))$ and $f(\eta(t))$ make an angle of θ' at $t = 0$ with $|\theta'| = |\theta|$ for all θ. Also, f is said to be **isotropic** if it magnifies lengths by some factor for all directions emanating from z_0, that is, if the limit

$$\lim_{r \to 0} \frac{|f(z_0 + re^{i\theta}) - f(z_0)|}{r}$$

exists, is non-zero, and independent of θ.

Then f is isogonal at z_0 if and only if it is isotropic at z_0; moreover, f is isogonal at z_0 if and only if either $f'(z_0)$ exists and is non-zero, or the same holds for f replaced by $\overline{f}$.

2. The angle between two non-zero complex numbers z and w (taken in that order) is simply the oriented angle, in $(-\pi, \pi]$, that is formed between the two vectors in $\mathbb{R}^2$ corresponding to the points z and w. This oriented angle, say α, is uniquely determined by the two quantities

$$\frac{(z,w)}{|z|\,|w|} \quad \text{and} \quad \frac{(z,-iw)}{|z|\,|w|}$$

which are simply the cosine and sine of α, respectively. Here, the notation $(\cdot,\cdot)$ corresponds to the usual Euclidian inner product in $\mathbb{R}^2$, which in terms of complex numbers takes the form $(z,w) = \mathrm{Re}(z\overline{w})$.

In particular, we may now consider two smooth curves $\gamma : [a,b] \to \mathbb{C}$ and $\eta : [a,b] \to \mathbb{C}$, that intersect at z_0, say $\gamma(t_0) = \eta(t_0) = z_0$, for some $t_0 \in (a,b)$. If the quantities $\gamma'(t_0)$ and $\eta'(t_0)$ are non-zero, then they represent the tangents to the curves γ and η at the point z_0, and we say that the two curves intersect at z_0 at the angle formed by the two vectors $\gamma'(t_0)$ and $\eta'(t_0)$.

A holomorphic function f defined near z_0 is said to **preserve angles** at z_0 if for any two smooth curves γ and η intersecting at z_0, the angle formed between the curves γ and η at z_0 equals the angle formed between the curves $f \circ \gamma$ and $f \circ \eta$ at $f(z_0)$. (See Figure 12 for an illustration.) In particular, we assume that the tangents to the curves γ, η, $f \circ \gamma$, and $f \circ \eta$ at the point z_0 and $f(z_0)$ are all non-zero.



Figure 12. Preservation of angles at z_0

(a) Prove that if $f : \Omega \to \mathbb{C}$ is holomorphic, and $f'(z_0) \neq 0$, then f preserves angles at z_0. [Hint: Observe that

$$(f'(z_0)\gamma'(t_0), f'(z_0)\eta'(t_0)) = |f'(z_0)|^2(\gamma'(t_0),\eta'(t_0)).]$$

(b) Conversely, prove the following: suppose $f : \Omega \to \mathbb{C}$ is a complex-valued function, that is real-differentiable at $z_0 \in \Omega$, and $J_f(z_0) \neq 0$. If f preserves angles at z_0, then f is holomorphic at z_0 with $f'(z_0) \neq 0$.

3.* The Schwarz-Pick lemma (see Exercise 13) is the infinitesimal version of an important observation in complex analysis and geometry.

For complex numbers $w \in \mathbb{C}$ and $z \in \mathbb{D}$ we define the **hyperbolic length** of w at z by

$$\|w\|_z = \frac{|w|}{1 - |z|^2},$$

where $|w|$ and $|z|$ denote the usual absolute values. This length is sometimes referred to as the **Poincaré metric**, and as a Riemann metric it is written as

$$ds^2 = \frac{|dz|^2}{(1 - |z|^2)^2}.$$

The idea is to think of w as a vector lying in the tangent space at z. Observe that for a fixed w, its hyperbolic length grows to infinity as z approaches the boundary of the disc. We pass from the infinitesimal hyperbolic length of tangent vectors to the global hyperbolic distance between two points by integration.

(a) Given two complex numbers z_1 and z_2 in the disc, we define the **hyperbolic distance** between them by

$$d(z_1, z_2) = \inf_{\gamma} \int_0^1 \|\gamma'(t)\|_{\gamma(t)}\, dt,$$

where the infimum is taken over all smooth curves $\gamma : [0, 1] \to \mathbb{D}$ joining z_1 and z_2. Use the Schwarz-Pick lemma to prove that if $f : \mathbb{D} \to \mathbb{D}$ is holomorphic, then

$$d(f(z_1), f(z_2)) \le d(z_1, z_2) \qquad \text{for any } z_1, z_2 \in \mathbb{D}.$$

In other words, holomorphic functions are distance-decreasing in the hyperbolic metric.

(b) Prove that automorphisms of the unit disc preserve the hyperbolic distance, namely

$$d(\varphi(z_1),\varphi(z_2)) = d(z_1, z_2), \qquad \text{for any } z_1, z_2 \in \mathbb{D}$$

and any automorphism φ. Conversely, if $\varphi : \mathbb{D} \to \mathbb{D}$ preserves the hyperbolic distance, then either φ or $\overline{\varphi}$ is an automorphism of $\mathbb{D}$.

(c) Given two points $z_1, z_2 \in \mathbb{D}$, show that there exists an automorphism φ such that $\varphi(z_1) = 0$ and $\varphi(z_2) = s$ for some s on the segment $[0, 1)$ on the real line.

(d) Prove that the hyperbolic distance between 0 and $s \in [0, 1)$ is

$$d(0, s) = \frac{1}{2} \log \frac{1 + s}{1 - s}.$$

(e) Find a formula for the hyperbolic distance between any two points in the unit disc.

4.* Consider the group of matrices of the form

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

that satisfy the following conditions:

(i) a, b, c, and $d \in \mathbb{C}$,

(ii) the determinant of M is equal to 1,

(iii) the matrix M preserves the following hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$:

$$\langle Z, W \rangle = z_1 \overline{w}_1 - z_2 \overline{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. In other words, for all $Z, W \in \mathbb{C}^2$

$$\langle MZ, MW \rangle = \langle Z, W \rangle.$$

This group of matrices is denoted by $\mathrm{SU}(1,1)$.

(a) Prove that all matrices in $\mathrm{SU}(1,1)$ are of the form

$$\begin{pmatrix} a & b \\ \overline{b} & \overline{a} \end{pmatrix},$$

where $|a|^2 - |\, b|^2 = 1$. To do so, consider the matrix

$$J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and observe that $\langle Z, W \rangle = {}^t\!W J Z$, where ${}^t\!W$ denotes the conjugate transpose of W.

(b) To every matrix in $\mathrm{SU}(1,1)$ we can associate a fractional linear transformation

$$\frac{az + b}{cz + d}.$$

Prove that the group $\mathrm{SU}(1,1)/\{\pm 1\}$ is isomorphic to the group of automorphisms of the disc. [Hint: Use the following association.]

$$e^{2i\theta} \frac{z - \alpha}{1 - \overline{\alpha} z} \longrightarrow \begin{pmatrix} \frac{e^{i\theta}}{\sqrt{1-|\alpha|^2}} & -\frac{\alpha e^{i\theta}}{\sqrt{1-|\alpha|^2}} \\ -\frac{\overline{\alpha} e^{-i\theta}}{\sqrt{1-|\alpha|^2}} & \frac{e^{-i\theta}}{\sqrt{1-|\alpha|^2}} \end{pmatrix}.$$

5. The following result is relevant to Problem 4 in Chapter 10 which treats modular functions.

(a) Suppose that $F : \mathbb{H} \to \mathbb{C}$ is holomorphic and bounded. Also, suppose that $F(z)$ vanishes when $z = ir_n$, $n = 1, 2, 3, \ldots$, where $\{r_n\}$ is a bounded sequence of positive numbers. Prove that if $\sum_{n=1}^{\infty} r_n = \infty$, then $F = 0$.

(b) If $\sum r_n < \infty$, it is possible to construct a bounded function on the upper half-plane with zeros precisely at the points ir_n.

For related results in the unit disc, see Problems 1 and 2 in Chapter 5.]

6.* The results of Exercise 18 extend to the case when γ is assumed merely to be closed, simple, and continuous. The proof, however, requires further ideas.

7.* Applying ideas of Carathéodory, Koebe gave a proof of the Riemann mapping theorem by constructing (more explicitly) a sequence of functions that converges to the desired conformal map.

Starting with a Koebe domain, that is, a simply connected domain $\mathcal{K}_0 \subset \mathbb{D}$ that is not all of $\mathbb{D}$, and which contains the origin, the strategy is to find an injective function f_0 such that $f_0(\mathcal{K}_0) = \mathcal{K}_1$ is a Koebe domain "larger" than $\mathcal{K}_0$. Then, one iterates this process, finally obtaining functions $F_n = f_n \circ \cdots \circ f_0 : \mathcal{K}_0 \to \mathbb{D}$ such that $F_n(\mathcal{K}_0) = \mathcal{K}_{n+1}$ and $\lim F_n = F$ is a conformal map from $\mathcal{K}_0$ to $\mathbb{D}$.

The **inner radius** of a region $\mathcal{K} \subset \mathbb{D}$ that contains the origin is defined by $r_{\mathcal{K}} = \sup\{\rho \geq 0 : D(0,\rho) \subset \mathcal{K}\}$. Also, a holomorphic injection $f : \mathcal{K} \to \mathbb{D}$ is said to be an **expansion** if $f(0) = 0$ and $|f(z)| > |z|$ for all $z \in \mathcal{K} - \{0\}$.

(a) Prove that if f is an expansion, then $r_{f(\mathcal{K})} \geq r_{\mathcal{K}}$ and $|f'(0)| > 1$. [Hint: Write $f(z) = zg(z)$ and use the maximum principle to prove that $|f'(0)| = |g(0)| > 1$.]

Suppose we begin with a Koebe domain $\mathcal{K}_0$ and a sequence of expansions $\{f_0, f_1, \ldots, f_n, \ldots\}$, so that $\mathcal{K}_{n+1} = f_n(\mathcal{K}_n)$ are also Koebe domains. We then define holomorphic maps $F_n : \mathcal{K}_0 \to \mathbb{D}$ by $F_n = f_n \circ \cdots \circ f_0$.

(b) Prove that for each n, the function F_n is an expansion. Moreover, $F_n'(0) = \prod_{k=0}^{n} f_k'(0)$, and conclude that $\lim_{n\to\infty} |f_n'(0)| = 1$. [Hint: Prove that the sequence $\{|F_n'(0)|\}$ has a limit by showing that it is bounded above and monotone increasing. Use the Schwarz lemma.]

(c) Show that if the sequence is osculating, that is, $r_{\mathcal{K}_n} \to 1$ as $n \to \infty$, then $\{F_n\}$ converges uniformly on compact subsets of $\mathcal{K}_0$ to a conformal map $F : \mathcal{K}_0 \to \mathbb{D}$. [Hint: If $r_{F(\mathcal{K}_0)} \geq 1$ then F is surjective.]

To construct the desired osculating sequence we shall use the automorphisms $\psi_\alpha = (\alpha - z)/(1 - \overline{\alpha}z)$.

(d) Given a Koebe domain $\mathcal{K}$, choose a point $\alpha \in \mathbb{D}$ on the boundary of $\mathcal{K}$ such that $|\alpha| = r_{\mathcal{K}}$, and also choose $\beta \in \mathbb{D}$ such that $\beta^2 = \alpha$. Let S denote the square root of ψ_α on $\mathcal{K}$ such that $S(0) = 0$. Why is such a function well defined? Prove that the function $f : \mathcal{K} \to \mathbb{D}$ defined by $f(z) = \psi_\beta \circ S \circ \psi_\alpha$

is an expansion. Moreover, show that $|f'(0)| = (1 + r_{\mathcal{K}})/2\sqrt{r_{\mathcal{K}}}$. [Hint: To prove that $|f(z)| > |z|$ on $\mathcal{K} - \{0\}$ apply the Schwarz lemma to the inverse function, namely $\psi_\alpha \circ g \circ \psi_\beta$ where $g(z) = z^2$.]

(e) Use part (d) to construct the desired sequence.

8.* Let f be an injective holomorphic function in the unit disc, with $f(0) = 0$ and $f'(0) = 1$. If we write $f(z) = z + a_2 z^2 + a_3 z^3 \cdots$, then Problem 1 in Chapter 3 shows that $|a_2| \leq 2$. Bieberbach conjectured that in fact $|a_n| \leq n$ for all $n \geq 2$; this was proved by deBranges. This problem outlines an argument to prove the conjecture under the additional assumption that the coefficients a_n are real.

(a) Let $z = re^{i\theta}$ with $0 < r < 1$, and show that if $v(r, \theta)$ denotes the imaginary part of $f(re^{i\theta})$, then

$$a_n r^n = \frac{2}{\pi} \int_0^\pi v(r, \theta) \sin n\theta \, d\theta.$$

(b) Show that for $0 \leq \theta \leq \pi$ and $n = 1, 2, \ldots$ we have $|\sin n\theta| \leq n \sin \theta$.

(c) Use the fact that $a_n \in \mathbb{R}$ to show that $f(\mathbb{D})$ is symmetric with respect to the real axis, and use this fact to show that f maps the upper half-disc into either the upper or lower part of $f(\mathbb{D})$.

(d) Show that for r small,

$$v(r, \theta) = r \sin \theta [1 + O(r)],$$

and use the previous part to conclude that $v(r, \theta) \sin \theta \geq 0$ for all $0 < r < 1$ and $0 \leq \theta \leq \pi$.

(e) Prove that $|a_n r^n| \leq nr$, and let $r \to 1$ to conclude that $|a_n| \leq n$.

(f) Check that the function $f(z) = z/(1 - z)^2$ satisfies all the hypotheses and that $|a_n| = n$ for all n.

9.* Gauss found a connection between elliptic integrals and the familiar operations of forming arithmetic and geometric means.

We start with any pair (a, b) of numbers that satisfy $a \geq b > 0$, and form the arithmetic and geometric means of a and b, that is,

$$a_1 = \frac{a+b}{2} \qquad \text{and} \qquad b_1 = (ab)^{1/2}.$$

We then repeat these operations with a and b replaced by a_1 and b_1. Iterating this process provides two sequences $\{a_n\}$ and $\{b_n\}$ where a_{n+1} and b_{n+1} are the arithmetic and geometric means of a_n and b_n, respectively.

(a) Prove that the two sequences $\{a_n\}$ and $\{b_n\}$ have a common limit. This limit, which we denote by $M(a, b)$, is called the **arithmetic-geometric mean** of a and b. [Hint: Show that $a \geq a_1 \geq a_2 \geq \cdots \geq a_n \geq b_n \geq \cdots \geq b_1 \geq b$ and $a_n - b_n \leq (a-b)/2^n$.]

(b) Gauss's identity states that

$$\frac{1}{M(a,b)} = \frac{2}{\pi} \int_0^{\pi/2} \frac{d\theta}{(a^2 \cos^2\theta + b^2 \sin^2\theta)^{1/2}}.$$

To prove this relation, show that if $I(a, b)$ denotes the integral on the right-hand side, then it suffices to establish the invariance of I, namely

$$I(a,b) = I\left(\frac{a+b}{2}, (ab)^{1/2}\right). \tag{6}$$

Then, observe that the connection with elliptic integrals takes the form

$$I(a,b) = \frac{1}{a}K(k) = \frac{1}{a}\int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \qquad \text{where } k^2 = 1 - b^2/a^2,$$

and that the relation (6) is a consequence of the identity in (b) of Exercise 24.

Chapter 9. An Introduction to Elliptic Functions

1. Besides the approach in Section 1.2, there are several alternate ways of dealing with the sum $\sum 1/(z+\omega)^2$, where $\omega = n + m\tau$. For example, one may sum either (a) circularly, (b) first in n then in m, (c) or first in m then in n.

(a) Prove that if $z \notin \Lambda$, then

$$\lim_{R\to\infty} \sum_{n^2+m^2\le R^2} \frac{1}{(z+n+m\tau)^2} = S_1(z)$$

exists and $S_1(z) = \wp(z) + c_1$.

(b) Similarly,

$$\sum_m \left(\sum_n \frac{1}{(z+n+m\tau)^2} \right) = S_2(z)$$

exists and $S_2(z) = \wp(z) + c_2$, where $c_2 = F(\tau)$, and F is the forbidden Eisenstein series.

(c) Also

$$\sum_n \left(\sum_m \frac{1}{(z+n+m\tau)^2} \right) = S_3(z)$$

exists with $S_3(z) = \wp(z) + c_3$, and $c_3 = \tilde{F}(\tau)$, the reverse of F.

[Hint: To prove (a), it suffices to show that $\lim_{R\to\infty}, \sum_{1\le n^2+m^2\le R^2} 1/(n+m\tau)^2 = c_1$ exists. This is proved by a comparision with $\int_{1\le x^2+y^2\le R^2} \frac{dx}{(x+y\tau)^2} = I(R)$. It can be shown that $I(R) = 0$, which follows because $(x+y\tau)^{-2} = -(\partial/\partial x\,)(x+y\tau)^{-1}$.]

2. Show that

$$\wp(z) = c + \pi^2 \sum_{m=-\infty}^{\infty} \frac{1}{\sin^2((z+m\tau)\pi)}$$

where c is an appropriate constant. In fact, by part (b) of the previous problem $c = -F(\tau)$.

3.* Suppose Ω is a simply connected domain that excludes the three roots of the polynomial $4z^3 - g_2 z - g_3$. For $\omega_0 \in \Omega$and ω_0 fixed, define the function I on Ω by

$$I(\omega) = \int_{\omega_0}^{\omega} \frac{dz}{\sqrt{4z^3 - g_2 z - g_3}} \qquad \omega \in \Omega.$$

Then the function I has an inverse given by $\wp(z + \alpha)$ for some constant α; that is,

$$I(\wp(z + \alpha)) = z$$

for appropriate α.

[Hint: Prove that $(I(\wp(z + \alpha)))' = \pm 1$, and use the fact that $\wp$ is even.]

4.* Suppose τ is purely imaginary, say $\tau = it$ with $t > 0$. Consider the division of the complex plane into congruent rectangles obtained by considering the lines $x = n/2$, $y = tm/2$ as n and m range over the integers. (An example is the rectangle whose vertices are $0, 1/2, 1/2 + \tau/2$, and $\tau/2$.)

(a) Show that $\wp$ is real-valued on all these lines, and hence on the boundaries of all these rectangles.

(b) Prove that $\wp$ maps the interior of each rectangle conformally to the upper (or lower) half-plane.

Chapter 10. Applications of Theta Functions

1.* Suppose n is of the form $n = 4^a(8k + 7)$, where a and k are positive integers. Show that n cannot be written as the sum of three-squares. The converse, that every n that is *not* of that form *can* be written as the sum of three-squares, is a difficult theorem of Legendre and Gauss.

2. Let $\mathrm{SL}_2(\mathbb{Z})$ denote the set of 2×2 matrices with integer entries and determinant 1, that is,

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \text{ and } ad - bc = 1 \right\}.$$

This group acts on the upper half-plane by the fractional linear transformation $g(\tau) = (a\tau + b)/(c\tau + d)$. Together with this action comes the so-called fundamental domain $\mathcal{F}_1$ in the complex plane defined by

$$\mathcal{F}_1 = \{\tau \in \mathbb{C} : |\tau| \geq 1, \quad |\mathrm{Re}(\tau)| \leq 1/2 \text{ and } |\mathrm{Im}(\tau)| \geq 0\}.$$

It is illustrated in Figure 3.

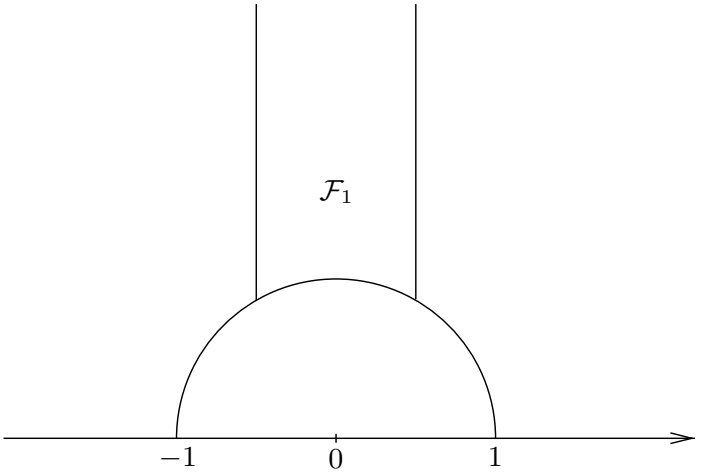


Figure 3. The fundamental domain $\mathcal{F}_1$

Consider the two elements in $\mathrm{SL}_2(\mathbb{Z})$ defined by $S(\tau) = -1/\tau$ and $T_1(\tau) = \tau + 1$. These correspond (for example) to the matrices

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

respectively. Let $\mathfrak{g}$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ generated by S and T_1.

(a) Show that for every $\tau \in \mathbb{H}$ there exists $g \in \mathfrak{g}$ such that $g(\tau) \in \mathcal{F}_1$.

(b) We say that two points τ and τ' are congruent if there exists $g \in \mathrm{SL}_2(\mathbb{Z})$ such that $g(\tau) = w$. Prove that if $\tau, w \in \mathcal{F}_1$ are congruent, then either $\mathrm{Re}(\tau) = \pm 1/2$ and $\tau' = \tau \mp 1$ or $|\tau| = 1$ and $\tau' = -1/\tau$. [Hint: Say $\tau' = g(\tau)$. Why can one assume that $\mathrm{Im}(\tau') \geq \mathrm{Im}(\tau)$, and therefore $|c\tau + d| \leq 1$? Now consider separately the possibilities $c = -1$, $c = 0$, or $c = 1$.]

(c) Prove that S and T_1 generate the modular group in the sense that every fractional linear transformation corresponding to $g \in \mathrm{SL}_2(\mathbb{Z})$ is a composition of finitely many S's and T_1's, and their inverses. Strictly speaking, the matrices associated to S and T_1 generate the projective special linear group $\mathrm{PSL}_2(\mathbb{Z})$, which equals $\mathrm{SL}_2(\mathbb{Z})$ modulo $\pm I$. [Hint: Observe that $2i$ is in the interior of $\mathcal{F}_1$. Now map $g(2i)$ back into $\mathcal{F}_1$ by using part (a). Use part (b) to conclude.]

3. In this problem, consider the group G of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with integer entries, determinant 1, and such that a and d have the same parity, b and c have the same parity, and c and d have opposite parity. This group also acts on the upper half-plane by fractional linear transformations. To the group G corresponds the fundamental domain $\mathcal{F}$ defined by $|\tau| \geq 1$, $|\mathrm{Re}(\tau)| \leq 1$, and $\mathrm{Im}(\tau) \geq 0$ (see Figure 1). Also, let

$$S(\tau) = -1/\tau \leftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad T_2(\tau) = \tau + 2 \leftrightarrow \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Prove that every fractional linear transformation corresponding to $g \in G$ is a composition of finitely many S, T_2 and their inverses, in analogy with the previous problem.

4. Let G denote the group of matrices given in the previous problem. Here we give an alternate proof of Theorem 3.4, that states that a function in $\mathbb{H}$ which is holomorphic, bounded, and invariant under G must be constant.

(a) Suppose that $f : \mathbb{H} \to \mathbb{C}$ is holomorphic, bounded, and that there exists a sequence of complex numbers $\tau_k = x_k + iy_k$ such that

$$f(\tau_k) = 0, \qquad \sum_{k=1}^{\infty} y_k = \infty, \qquad 0 < y_k \leq 1, \qquad \text{and} \qquad |x_k| \leq 1.$$

Then $f = 0$. [Hint: When $x_k = 0$ see Problem 5 in Chapter 8.]

(b) Given two relatively prime integers c and d with different parity, show that there exist integers a and b such that $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$. [Hint: All the solutions of $xc + dy = 1$ take the form $x_0 + dt$ and $y_0 - ct$ where x_0, y_0 is a particular solution and $t \in \mathbb{Z}$.]

(c) Prove that $\sum 1/(c^2 + d^2) = \infty$ where the sum is taken over all c and d that are relatively prime and of opposite parity. [Hint: Suppose not, and prove that $\sum_{(a,b)=1} 1/(a^2 + b^2) < \infty$ where the sum is over all relatively prime integers a and b. To do so, note that if a and b are both odd and relatively prime, then the two numbers c and d defined by $c = (a + b)/2$ and $d = (a - b)/2$ are relatively prime and of opposite parity. Moreover, $c^2 + d^2 \leq A(a^2 + b^2)$ for some universal constant A. Therefore

$$\sum_{n \neq 0} \frac{1}{n^2} \sum_{(a,b)=1} \frac{1}{a^2 + b^2} < \infty,$$

hence $\sum 1/(k^2 + \ell^2) < \infty$, where the sum is over all integers k and ℓ such that $k, \ell \neq 0$. Why is this a contradiction?]

(d) Prove that if $F : \mathbb{H} \to \mathbb{C}$ is holomorphic, bounded, and invariant under G, then F is constant. [Hint: Replace $F(\tau)$ by $F(\tau) - F(i)$ so that we can assume $F(i) = 0$ and prove $F = 0$. For each relatively prime c and d with opposite parity, choose $g \in G$ so that $g(i) = x_{c,d} + i/(c^2 + d^2)$ with $|x_{c,d}| \leq 1$.]

5.* In Chapter 9 we proved that the Weierstrass $\wp$ function satisfies the cubic equation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3,$$

where $g_2 = 60E_4$, $g_3 = 140E_6$, with E_k is the Eisenstein series of order k. The discriminant of the cubic $y^2 = 4x^3 - g_2x - g_3$ is defined by $\triangle = g_2^3 - 27g_3^2$. Prove that

$$\triangle(\tau) = (2\pi)^{12}\eta^{24}(\tau) \qquad \text{for all } \tau \in \mathbb{H}.$$

[Hint: $\triangle$ and η^{24} satisfy the same transformation laws under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$. Because of the fundamental domain described in Problem 2, it suffices then to investigate the behavior at the only cusp, which is at infinity.]

6.* Here we will deduce the formula for $r_8(n)$, which counts the number of representations of n as a sum of eight squares. The method is parallel to that of $r_4(n)$, but the details are less delicate.

Theorem: $r_8(n) = 16\sigma_3^*(n)$.

Here $\sigma_3^*(n) = \sigma_3(n) = \sum_{d|n} d^3$, when n is odd. Also, when n is even

$$\sigma_3^*(n) = \sum_{d|n} (-1)^d d^3 = \sigma_3^e(n) - \sigma_3^o(n),$$

where $\sigma_3^e(n) = \sum_{d|n,\ d \text{ even}} d^3$ and $\sigma_3^o(n) = \sum_{d|n,\ d \text{ odd}} d^3$.

Consider the appropriate Eisenstein series

$$E_4^*(\tau) = \sum \frac{1}{(n+m\tau)^4},$$

where the sum is over integers n and m with opposite parity. Recall the standard Eisenstein series

$$E_4(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^4}.$$

Notice that the series defining E_4^* is absolutely convergent, in distinction to $E_2^*(\tau)$, which arose when considering $r_4(n)$. This makes some of the considerations below quite a bit simpler.

(a) Prove that $r_8(n) = 16\sigma_3^*(n)$ is equivalent to the identity $\theta(\tau)^8 = 48\pi^{-4}E_4^*(\tau)$. [Hint: Use the fact that $E_4(\tau) = 2\zeta(4) + \frac{(2\pi)^4}{3}\sum_{k=1}^{\infty}\sigma_3(k)e^{2\pi ik\tau}$ and $\zeta(4) = \pi^4/90$.]

(b) Note that $E_4^*(\tau) = E_4(\tau) - 2^{-4}E_4((\tau-1)/2)$.

(c) $E_4^*(\tau+2) = E_4^*(\tau)$.

(d) $E_4^*(\tau) = \tau^{-4}E_4^*(-1/\tau)$.

(e) $(48/\pi^4)E_4^*(\tau) \to 1$ as $\tau \to \infty$.

(f) $|E_4^*(1-1/\tau)| \approx |\ \tau|^4 |e^{2\pi i\tau}|$, as $\mathrm{Im}(\tau) \to \infty$. [Hint: Verify that $E_4^*(1-1/\tau) = \tau^4(E_4(\tau) - E_4(2\tau))$.]

Since $\theta(\tau)^8$ satisfies properties similar to (c), (d), (e) and (f) above, it follows that the invariant function $48\pi^{-4}E_4^*(\tau)/\theta(\tau)^8$ is bounded and hence a constant, which must be 1. This gives the desired result.

Appendix A. Asymptotics

1. Let η be an indefinitely differentiable function supported in a finite interval, so that $\eta(x) = 1$ for x near 0. Then, if $m > -1$ and $N > 0$,

$$\int_0^\infty e^{-sx} x^m \eta(x)\, dx = s^{-m-1}\Gamma(m+1) + O(s^{-N})$$

for $\mathrm{Re}(s) \geq 0$, $|s| \to \infty$.

(a) Consider first the case $-1 < m \leq 0$. It suffices to see that

$$\int_0^\infty e^{-sx} x^m (1 - \eta(x))\, dx = O(s^{-N}),$$

and this can be done by repeated integration by parts since $e^{-sx} = (-1)^N s^{-N} \left(\frac{d}{dx}\right)^N (e^{-sx})$.

(b) To extend this to all m, find an integer k so that $k - 1 < m \leq k$, write

$$\int \left[\left(\frac{d}{dx} \right)^k (x^m) \right] e^{-sx} \eta(x)\, dx = c_{k,m} s^{-m+k-1} + O(s^{-N}),$$

and integrate by parts k times.

2. The following is a more precise version of Stirling's formula. There are real constant $a_1 = 1/12$, a_2, ..., a_n, ..., so that for every $N > 0$

$$\Gamma(s) = e^{s \log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \left(1 + \sum_{j=1}^{N} a_j s^{-j} + O(s^{-N}) \right) \qquad \text{when } s \in S_\delta.$$

This can be proved by using the results of Problem 1 in place of Proposition 1.3.

3. The Bessel functions and Airy function have the following power series expansions:

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{x^2}{4}\right)^m}{m!\Gamma(\nu+m+1)},$$

$$\mathrm{Ai}(-x) = \frac{1}{\pi} \sum_{n=0}^{\infty} \frac{x^n}{n!} \sin(2\pi(n+1)/3) 3^{n/3-2/3} \Gamma(n/3+1/3).$$

(a) From this, verify that when $x > 0$,

$$\mathrm{Ai}(-x) = \frac{x^{1/2}}{3}\left(J_{1/3}\left(\frac{2}{3}x^{3/2}\right) + J_{-1/3}\left(\frac{2}{3}x^{3/2}\right)\right).$$

(b) The function $\mathrm{Ai}(x)$ extends to an entire function of order $3/2$.

[Hint: For (b), use (a), or alternatively, apply Problem 4 in Chapter 5 to the power series for Ai. Compare also with Problem 1, Chapter 4.]

4. Suppose $F(z) = \sum_{n=0}^{\infty} F_n w^n$ is meromorphic in a region containing the closed unit disc, and the only poles of F are on the unit circle at the points $\alpha_1, \ldots, \alpha_k$, and their orders are $r_1, \ldots, r_k$ respectively. Then for some $\epsilon > 0$

$$F_n = \sum_{j=1}^{k} P_j(n) + O(e^{-\epsilon n}) \qquad \text{as } n \to \infty.$$

Here

$$P_j(n) = \frac{1}{(r_j - 1)!} \left(\frac{d}{dw} \right)^{r_j - 1} \left[(w - \alpha_j)^{r_j} w^{-n-1} F(w) \right]_{w=\alpha_j}.$$

Note that each P_j is of the form $P_j(n) = A_j(\alpha_j^{-1} n)^{r_j - 1} + O(n^{r_j - 2})$.

To prove this, use the residue formula (Theorem 1.4, Chapter 3).

5.* The one shortcoming in our derivation of the asymptotic formula for $p(n)$ arose from the fact that while $f_1(z) = \sqrt{z/i}\ e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}}$ is a good approximation to the generating function $f(z)$ near $z = 0$, this fails near other points on the real axis, since f_1 is regular there, but f is not.

However, using the transformation law (24) and the identity $f(z+1) = f(z)$, one can derive the following generalization of (24): whenever p/q is a rational number in lowest form (so p and q are relatively prime) then

$$f\left(z - \frac{p}{q}\right) = \omega_{p/q}\sqrt{\frac{zq}{i}}\ e^{\frac{i\pi}{12zq^2}} e^{-\frac{i\pi z}{12}} f\left(-\frac{1}{zq^2} - \frac{p'}{q}\right),$$

where $pp' = 1 \mod q$. Here $\omega_{p/q}$ is an appropriate 24th root of unity. This formula leads to an analogous $f_{p/q}$, approximating f at $z = p/q$.

From this one can obtain for each p/q a contribution of the form

$$c_{p/q}\frac{1}{2\pi\sqrt{2}}\frac{d}{dn}\left(\frac{e^{\frac{K}{q}(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}}\right)$$

to the asymptotic formula for $p(n)$. When suitably modified, the resulting series, summed over all proper fractions p/q in $[0, 1)$, actually converges and gives an exact formula for $p(n)$.